Pneumonia

ORG: M-282 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
28th Edition

Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Admission to Inpatient Care

Note: Some patients may be appropriate for Observation care. For consideration of observation care, see Pneumonia: Observation Care ISC.

Note: For patients with clinically active secondary conditions, see Pneumonia Multiple Condition Management Guidelines.

- Admission is indicated for **1 or more** of the following[A][B](1)(2)(5)(6)(7)(8):
  - Hypoxemia
  - Hemodynamic instability
  - Altered mental status that is severe or persistent
  - Dehydration that is severe or persistent
  - Ventilatory assistance needed (eg, mechanical ventilation, noninvasive ventilation)
  - Bacteremia (if blood cultures performed)
  - Moderate-risk-category or high-risk-category patient[C] (Pneumonia Severity Index class IV or V, or CURB-65 score of 3 or greater) PSI Calculator CURB-65 Calculator
  - Respiratory finding (eg, Tachypnea) that persists despite observation care
  - Complicated pleural effusions (eg, empyema, loculated)
  - Presence of risk factor for poor outcome (eg, gross hemoptysis, cavitary infiltrate, neuromuscular weakness, cystic fibrosis)
  - Isolation indicated that cannot be performed outside hospital setting

Alternatives to Admission

- Alternatives include(1)(2)(5)(7)(8)(12)(13):
  - Outpatient care in emergency department, rapid treatment site, urgent care center, or medical office
    - History, physical examination, and assessment of risk factors
    - Chest x-ray and laboratory testing

        - Antibiotics and re-evaluation of hemodynamics, oxygenation status, and other parameters
        - Discharge on oral or parenteral antibiotics.
    - Observation. See Pneumonia: Observation Care ISC guideline as appropriate.
    - Home care(14)
        - Nursing visits for assessment and laboratory work
        - Parenteral antibiotics, respiratory therapy, and oxygen, as indicated
    - Recovery facility
        - Parenteral fluid and medication
        - Respiratory treatment and oxygen
        - Frequent assessment and treatment
        - Management of comorbidities

Alternative Care Planning

- Care planning needs for patient not requiring admission may include(15)(16):
    - Diagnostic tests, including(1)(2)(5)(7)(8)(12):
        - Pre-antibiotic sputum culture and Gram stain(17)
        - Pre-antibiotic blood culture
        - Chest x-ray
        - Chest CT
        - Chest ultrasound
        - Pulse oximetry
        - ABG
        - CBC
        - Chemistry panel
        - C-reactive protein(12)(15)
        - Urinary antigen tests (eg, for *Streptococcus pneumoniae* and *Legionella* species)(18)
        - Nasopharyngeal, nasal, or oropharyngeal swab for antigen or nucleic acid amplification test (eg, PCR) for viral pathogens(19)(20)(21)(22)
        - Sputum nucleic acid amplification test (eg, PCR) for *Legionella* species(18)
        - HIV serology
        - Serum serology (eg, *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*)
    - Treatments and procedures, including:
        - Antibiotics(18)(23)
        - Antiviral medication (eg, if proven or suspected viral co-infection)(21)
        - Respiratory therapy
    - Discharge Planning as appropriate
    - Patient, family, and caregiver education as appropriate. See Pneumonia: Patient Education for Clinicians SR.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	• ICU[D] or floor • Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment • Discharge planning	• **Clinical Indications met[E]** • Possible Fever, dyspnea, purulent sputum, pleuritic pain, Altered mental status	• Activity as tolerated	• Possible IV fluids • Parenteral or oral medications • Liquid or usual diet	• WBC • Possible ABG or oximetry • Blood culture • Possible sputum Gram stain and culture • Possible respiratory therapy • Probable oxygen • Possible thoracentesis • Incentive spirometry • Head of bed at 30 degrees	• IV antibiotics

2	• Floor • Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment	• **No CO_2 retention or acidosis** • **No requirement for mechanical ventilation** • **Hypotension absent** • **Afebrile or fever improved** • **No hypoxia on room air or oxygenation improved** • **Mental status improved or at baseline**	• **Increased activity**	• Oral hydration • Oral medications • Usual diet	• Incentive spirometry • Pulse oximetry • Head of bed at 30 degrees • Possible oxygen	• IV or oral antibiotics
3	• Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment • Floor to discharge[F] • Complete discharge planning	• **Hemodynamic stability** • **Afebrile or temperature acceptable for next level of care** • **Tachypnea absent** • **Hypoxemia absent** • **Mental status at baseline** • **Antibiotic regimen acceptable for next level of care** • **Discharge plans and education understood**	• **Ambulatory or acceptable for next level of care**	• **Oral hydration[G]** • **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**	• **Oxygen absent or at baseline need** • **Isolation not indicated, or is performable at next level of care** • WBC • Incentive spirometry	• Oral antibiotics

(2)(5)(7)(8)(10)(12)(26)(27)(28)

Care Task List

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 2 days

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Note: See Pneumonia Multiple Condition Management Benchmarking Table for more detailed information.

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(2)(7)(8)(10):
  - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
    - Expect brief stay extension.
  - Unclear diagnosis
    - Patient with negative cultures who is not recovering (eg, persistent signs and symptoms) on empiric antibiotics may require bronchoscopy, open lung biopsy, pleural biopsy, or changed antibiotic regimen.
    - Expect brief stay extension.
  - Pleural disease

    - Large pleural effusions or empyema may require repetitive drainage after diagnostic thoracentesis, chest tube drainage, or video-assisted thoracoscopy.
    - Expect brief stay extension.
  - Severe pneumonia or treatment failure(24)(29)
    - Nonresponsiveness to initial therapy has been associated with higher initial severity of infection (eg, high Pneumonia Severity Index or CURB-65 scores), hypoxemia, respiratory rate greater than 30, and thrombocytopenia.(29)
    - Patient with necrotizing pneumonia or lung abscess may require longer hospital stay for recovery.(30)
    - Patient with extension of x-ray infiltrates, multilobar disease, or ongoing hypoxemia may require longer hospital stay for recovery.
    - Patient with bacterial and viral co-infection may have higher morbidity and require longer hospital stay.(21)
    - Expect brief stay extension.
  - Infection with antibiotic-resistant organism (eg, *Pseudomonas* species, *Acinetobacter* species, methicillin-resistant *Staphylococcus aureus* (MRSA))[B][H][I](3)(4)(33)
    - Patient infected with antibiotic-resistant organism may require multiple antibiotics, broader coverage, and more prolonged IV antibiotic course.
    - Expect brief stay extension.
  - Respiratory insufficiency or failure
    - Anticipate invasive or noninvasive ventilatory support or high-flow nasal oxygen.(27)(34)
    - Expect moderate stay extension.
  - New-onset hyponatremia (serum sodium concentration of 135 mEq/L (mmol/L) or less)
    - Anticipate close monitoring for signs and symptoms and of serum electrolytes.
    - Expect brief stay extension.
  - Clinically significant comorbid illness (eg, heart failure, atrial fibrillation with rapid heart rate, alcohol withdrawal, renal insufficiency, diabetes)
    - Anticipate evaluation and treatment of specific comorbidity.
    - Expect brief stay extension.
  - Comorbid acute exacerbation of COPD(35)
    - COPD is associated with higher mortality, higher rates of ventilator-dependent respiratory failure, and *Pseudomonas* infection.
    - Expect brief stay extension.
  - Concomitant diagnosis of malignancy (eg, postobstruction pneumonia)
    - Malignancy may be associated with malnutrition, immunologic impairment, or bronchial obstruction.
    - Expect brief to moderate stay extension.
  - Altered mental status
    - Altered mental status disease may delay mobilization and recovery.
    - Expect brief stay extension.
  - Malnutrition(36)
    - Malnutrition is associated with higher mortality, higher rates of ventilator-dependent respiratory failure, and higher risk of sepsis and septic shock.
    - Expect brief stay extension.

See Common Complications and Conditions ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(2)(5)(7)(8)(12)(16)(18)(24)(27):
  - Diagnostic test scheduling and completion, including(44):
    - WBC count
    - Sputum Gram stain, culture(17)
    - Blood culture
    - Nasopharyngeal, nasal, or oropharyngeal swab for antigen or nucleic acid amplification test (eg, PCR) for viral pathogens(19)(20)(21)(22)
    - Nasal swab for MRSA[I](45)
    - Viral serologies
    - Nucleic acid amplification test (eg, PCR), mass spectrometry, or fluorescence in situ hybridization for respiratory pathogens(20)(46)
    - Sputum nucleic acid amplification test (eg, PCR) for *Legionella* species(18)
    - Urinary antigen tests (eg, for *Streptococcus pneumoniae* and *Legionella* species)(18)
    - Cultures, serologic testing, and fungal biomarkers for opportunistic infections for Immunosuppressed patients[J](44)
    - C-reactive protein or procalcitonin(12)(47)
    - ABG
    - Chest x-ray
    - Chest ultrasound

        - Lung ventilation perfusion scan
        - Chest CT scan
        - Echocardiogram
    - Treatment and procedure scheduling and completion, including:
        - IV antibiotics and switch to oral antibiotics(23)(44)
        - Antiviral medication (eg, if proven or suspected influenza or viral co-infection)(19)(21)
        - Nebulized bronchodilators
        - Systemic corticosteroids(2)(12)(37)(38)(39)
        - Incentive spirometry
        - Thoracentesis
        - Placement of parenteral access line
        - Bronchoscopy with bronchoalveolar lavage or biopsy (eg, for Immunosuppressed patients or nonresolving pneumonia)(44)
    - Consultation, assessment, and other services scheduling and completion, including:
        - Pulmonary medicine consultation
        - Infectious diseases consultation
        - Smoking cessation counseling
    - Identification of patient at high risk for readmission to prioritize transition and post-acute care:
        - Risk of readmission is increased by presence of **1 or more** of the following(42)(43)(48)(49)(50)(51)(52)(53):
            - Hospitalization (nonelective) in past 3 months
            - 2 or more emergency department visits in past 6 months
            - Admission from long-term care facility (eg, nursing home)
            - Prescribed antibiotics within 30 days prior to admission
            - Immunosuppressed patient
            - No source of outpatient care other than emergency department (eg, no primary care provider)(54)(55)(56)(57)(58)
            - Severe care transition barriers (eg, no caregiver, homeless)(56)(57)(58)(59)(60)(61)
            - Severe or end-stage renal disease (on dialysis or GFR less than 30 mL/min/1.73m^2 (0.5 mL/sec/1.73m^2))(59)(62)
              eGFR - Adult Calculator
    - Implementation of intervention that may reduce risk of readmission
        - Initial antibiotic treatment includes coverage of atypical organisms (eg, *Legionella pneumoniae*, *Chlamydophila pneumoniae*).(40)(41)
    - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool SR); key items include(63)(64):
        - Hemodynamic stability
        - Respiratory status, including oxygenation and need for respiratory therapy
        - Mental status
        - Signs and symptoms of sepsis
        - Nutritional status, including hydration

Discharge

Discharge Planning

- Discharge planning includes[K]:
    - Assessment of needs and planning for care, including(66)(67):
        - Develop and modify treatment plan (involving multiple providers) as needed.
        - Evaluate and address preadmission functioning as needed.
        - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
        - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(65)
        - Evaluate and address patient or caregiver preferences as indicated.
        - Identify skilled services needed at next level of care, with specific attention to:
            - Medication management, adherence instruction, and side effects assessment(68)(69)
            - Ongoing education required(70)
            - Respiratory status assessment
    - Early identification of anticipated discharge destination; options include(67)(71):
        - Home, considerations include:
            - Home safety assessment. See Home Safety Assessment SR for further information.
            - Patient safe to go home; examples include(72)(73)(74):
                - Medical status stable for patient's condition

- Functional care can safely be provided with available resources.
- Mental status stable for patient's condition
- Medication availability confirmed and reconciliation complete
- Patient/caregiver education completed with written discharge instructions provided
- Community resources identified and referrals made, as needed
- Home care arranged, if indicated
- Necessary medical equipment delivery arranged or available in home, if indicated
- Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated

- Access to follow-up care
- Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
- Caregiver need, ability, and availability

- Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.

- Transitions of care plan complete, including(67)(71)(75):
  - Patient and caregiver education complete. See Pneumonia: Patient Education for Clinicians SR for further information.
    - See Teach Back Tool SR for further information.
  - Medication reconciliation completion includes(69)(76):
    - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
    - Assess each medication for correlation to disease state or medical condition.
    - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
    - Provide reconciled medication list to all treating providers.
    - Confirm that patient or caregiver can acquire medication.
    - Educate patient and caregiver.
      - Provide complete medication list to patient and caregiver.
      - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
      - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(77)
    - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
    - See Medication Reconciliation Tool SR for further information.
  - Plan communicated to patient, caregiver, and all members of care team, including(78):
    - Inpatient care and service providers
    - Primary care provider
    - All post-discharge care and service providers
  - Appointments planned or scheduled, which may include:
    - Primary care provider
    - Follow-up evaluation call. See Post-Acute Care Follow-Up Call Tool SR for more information.(70)
    - Infectious disease specialist(79)
    - Pulmonologist
    - Respiratory therapy(70)
    - Other
  - Outpatient testing and procedure plans made, which may include(80):
    - Laboratory testing
    - Radiology(79)
    - Other
  - Referrals made for assistance or support, which may include:
    - Alcohol and other drug abuse or dependence treatment(79)
    - Community services
    - Financial, for follow-up care, medication, and transportation
    - Tobacco use treatment(79)
    - Other
  - Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
    - Incentive spirometer(70)
    - Nebulizer
    - Oxygen and supplies
    - Other

Discharge Destination

- Post-hospital levels of admission may include:
  - Home.
  - Home healthcare. See Home Care Indications for Admission Section HC in Pneumonia guideline in Home Care.
  - Recovery facility care. See Recovery Facility Care Indications for Admission Section RFC in Pneumonia guideline in Recovery Facility Care.

Evidence Summary

Background

For patients with pneumonia suspected or known to be of viral etiology other than COVID-19 (eg, positive test for influenza), the Viral Illness, Acute ISC guideline is appropriate. For viral infection due to COVID-19, the COVID-19 ISC guideline should be used. If the etiology is bacterial or unknown (eg, treating with antibiotics empirically), the Pneumonia guideline is appropriate.

This guideline pertains to the vast majority of patients with pneumonia, including patients deemed to have healthcare-associated pneumonia (HCAP). HCAP is defined as pneumonia developing in a patient with one or more of the following healthcare contacts: residence in a nursing home or long-term care facility, wound care, IV chemotherapy or hemodialysis within the previous 30 days, or acute care hospitalization for 2 or more days within the previous 90 days.(1) **(EG 2)** Concerns have been raised as to whether a patient identified as having HCAP is truly more likely to be infected with a multidrug-resistant organism.(2)(3)(4) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 3 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 3 book sections.

Analysis of national hospital discharge data for a commercially insured population shows 24.9% of patients seen in an emergency department with a principal diagnosis of pneumonia were admitted to inpatient care.(9) **(EG 3)** Analysis of national hospital discharge data for a Medicare-insured population shows 61.2% of patients seen in an emergency department with a principal diagnosis of pneumonia were admitted to inpatient care.(9) **(EG 3)** Analysis of an all-payer database shows a mean length of stay of 5.0 days for patients admitted to inpatient care.(10) **(EG 3)**

An emergency medicine textbook recommends that patients evaluated by the Pneumonia Severity Index (PSI) as high risk (class IV or V) should be admitted to inpatient care, that moderate-risk patients (class III) can be initially treated in observation care, and that, absent other indications for admission, low-risk patients (class I or II) can be treated as outpatients.(1) **(EG 2)** Clinical variables that are included in the PSI, and thus contribute to a higher risk score, are tachycardia, hypotension, altered mental status, tachypnea, and hypoxemia.(1) **(EG 2)** A specialty society guideline recommends use of the PSI in addition to clinical judgment in the evaluation of pneumonia patients and specifies that low-risk patients can be treated as outpatients.(2) **(EG 2)** This same guideline identifies major and minor criteria that would denote a patient as having severe pneumonia (inpatient treatment appropriate, likely ICU). Examples include respiratory failure (ventilatory assistance needed), hypotension, hypoxemia, and altered mental status.(2) **(EG 2)** Another specialty society guideline also recommends the use of PSI to help identify patients who require inpatient care, specifically outpatient care for low-risk patients (class I or II), observation care for moderate-risk patients (class III), and inpatient care for high-risk patients (class IV or V).(6) **(EG 2)**

Hospitalization

Systematic reviews and clinical guidelines found mixed evidence regarding the efficacy of corticosteroids as adjunctive therapy for severe community-acquired pneumonia.(2)(12)(37)(38) **(EG 2)** Current clinical guidelines recommend limiting their use to treatment of septic shock refractory to fluid resuscitation and for patients with concomitant conditions requiring their use (eg, asthma, COPD exacerbation).(2)(12) **(EG 2)** However, following publication of these guidelines and reviews, a randomized study of 795 patients admitted with severe community-acquired pneumonia and without septic shock (median age 67 years, 83% with baseline Pneumonia Severity Index class IV or V) found that early addition of hydrocortisone (within 24 hours of fulfilling severity criteria) reduced 28-day mortality (6.2% vs 11.9%, difference -5.6% (95% confidence interval (CI) -9.6% to -1.7%)), need for intubation (19.5% vs 27.7%, hazard ratio (HR) 0.59 (95% CI 0.40 to 0.86)), and need for new vasopressors (if not receiving vasopressor at study entry; 15.3% vs 25.0%, HR 0.59 (95% CI 0.43 to 0.82)).(39) **(EG 1)**

Readmission risk and reduction: An analysis of 2209 Medicare patients (mean age 77 years) treated for bacteremic community-acquired pneumonia found, after multivariate adjustment, that treatment with an initial antibiotic regimen that included coverage of atypical organisms (eg, macrolide, fluoroquinolone) was independently associated with a lower readmission rate.(40) **(EG 2)** A subsequent trial of 580 patients admitted for moderately severe community-acquired pneumonia (median age 76 years, 72% Pneumonia Severity Index class III or IV) and randomized to either a beta-lactam plus a macrolide or beta-lactam monotherapy found that the combination therapy group had fewer 30-day readmissions (3.1% vs 7.9%).(41) **(EG 1)** Analysis of a cohort of 977 patients admitted for bacterial pneumonia found that admission from a long-term care facility, immunosuppression (eg, malignancy, chemotherapy, systemic corticosteroids, AIDS), prior antibiotics within 30 days, and prior hospitalization within 90 days were independent predictors for all-cause 30-day readmission.(42) **(EG 2)** Database analysis of 45,134 pneumonia patients treated at a veteran's hospital who were age 65 years or older found, after multivariate analysis, that independent predictors of 30-day hospital readmission were admission from a nursing home and prior hospitalization within 90 days.(43) **(EG 2)**

Length of Stay

A retrospective study of 2169 adults (median age 66 years) admitted with pneumonia found that 25% were discharged within 2 days.(26) **(EG 2)** Examination of a cohort of 2320 adults (median age 57 years) hospitalized for community-acquired pneumonia with radiographic evidence of pneumonia found that 25% of patients were discharged in 2 days or less.(28) **(EG 2)** Analysis of national hospital discharge data shows 26% of hospitalized adult patients with a principal diagnosis of community-acquired pneumonia discharged in 2 days or less.(10) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify patient-specific complex clinical factors that make it reasonable to expect a necessity for hospital care across 2 or more midnights. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately admit a patient to inpatient care. For Medicare enrollees, MCG care guidelines also support the clinician in the decision to appropriately admit a patient to inpatient care when there is not an expectation of 2 or more midnights of hospital care (eg, CMS' Two-Midnight Rule exceptions).

Use of these evidence-based clinical criteria to support decision making around the need for inpatient treatment is of clinical benefit to the patient and is crucial for quality patient care. Use of evidence-based criteria ensures patients receive the most appropriate care for their specific condition, reduces unnecessary risks, promotes recovery, and provides a standard that can reduce disparities in care. Not only do evidence-based criteria help align treatment decisions with best practice, but they can also help reduce unwarranted variation in care by fostering consistency and equality in healthcare provision across regions and facilities. Appropriate use of the evidence-based clinical criteria in conjunction with the Supplemental Medicare Criteria ensures Medicare patients receive full access to Medicare benefits (eg, inpatient care), without risk of delay or decreased access, based on variables such as clinical severity of illness, length of provision of medically necessary hospital-level care, or satisfaction of specified Medicare criteria as directed by CMS. Use of these criteria will help ensure that the patient receives necessary care in the appropriate setting.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(11); 42 CFR 419.22(81); 42 CFR 422.101(82)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(83); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(84); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(85); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(86)

Medicare Coverage Determinations: Medicare Coverage Database(87)

References

1. Waxman MA, Moran GJ. Pneumonia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:829-838.e2. [Context Link 1, 2, 3, 4, 5, 6, 7]
2. Metlay JP, et al. Diagnosis and treatment of adults with community-acquired pneumonia. an Official Clinical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. American Journal of Respiratory and Critical Care Medicine 2019;200(7):e45-e67. DOI: 10.1164/rccm.201908-1581ST. (Reaffirmed 2023 Apr) [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22] View abstract...
3. Kalil AC, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. Clinical Infectious Diseases 2016;63(5):e61-e111. DOI: 10.1093/cid/ciw353. (Reaffirmed 2023 Jun) [Context Link 1, 2, 3] View abstract...
4. Ewig S, Kolditz M, Pletz MW, Chalmers J. Healthcare-associated pneumonia: is there any reason to continue to utilize this label in 2019? Clinical Microbiology and Infection 2019;25(10):1173-1179. DOI: 10.1016/j.cmi.2019.02.022. [Context Link 1, 2, 3] View abstract...
5. Klompas M. Healthcare and hospital-acquired pneumonia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1514-1518. [Context Link 1, 2, 3, 4, 5]
6. American College of Emergency Physicians Clinical Policies Subcommittee (Writing Committee) on Commu, et al. Clinical policy: critical issues in the management of adult patients presenting to the emergency department with community-acquired pneumonia. Annals of Emergency Medicine 2021;77(1):e1-e57. DOI: 10.1016/j.annemergmed.2020.10.024. [Context Link 1, 2, 3, 4] View abstract...
7. Musher DM. Community-acquired pneumonia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1503-1513. [Context Link 1, 2, 3, 4, 5, 6]

8. Aliberti S, Dela Cruz CS, Amati F, Sotgiu G, Restrepo MI. Community-acquired pneumonia. Lancet 2021;398(10303):906-919. DOI: 10.1016/S0140-6736(21)00630-9. [Context Link 1, 2, 3, 4, 5, 6] View abstract...
9. Proprietary health insurance data sources (2021-2022); and Medicare 5% Standard Analytical File (2020-2021). [Context Link 1, 2]
10. Premier PINC AI™ Healthcare Database (PHD), 01/01/2021-12/31/2022. Premier, Inc. [Context Link 1, 2, 3, 4]
11. Centers for Medicare and Medicaid Services. "Admissions." 42 CFR 412.3 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-412. [Context Link 1, 2]
12. Pneumonia in Adults: Diagnosis and Management. NICE Clinical Guideline CG191 [Internet] National Institute for Health and Care Excellence. 2022 Jul 07 Accessed at: http://guidance.nice.org.uk. [accessed 2023 Oct 11] [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9]
13. Pena ME, Kazan VM, Helmreich MN, Mace SE. Care of respiratory conditions in an observation unit. Emergency Medicine Clinics of North America 2017;35(3):625-645. DOI: 10.1016/j.emc.2017.03.008. [Context Link 1] View abstract...
14. Levine DM, et al. Hospital-level care at home for acutely Ill adults: a randomized controlled trial. Annals of Internal Medicine 2020;172(2):77-85. DOI: 10.7326/M19-0600. [Context Link 1] View abstract...
15. Hill AT, et al. Adult outpatients with acute cough due to suspected pneumonia or influenza: CHEST guideline and expert panel report. Chest 2019;155(1):155-167. DOI: 10.1016/j.chest.2018.09.016. [Context Link 1, 2] View abstract...
16. Torres A, et al. Pneumonia. Nature Reviews. Disease Primers 2021;7(1):25. DOI: 10.1038/s41572-021-00259-0. [Context Link 1, 2] View abstract...
17. Del Rio-Pertuz G, et al. Usefulness of sputum gram stain for etiologic diagnosis in community-acquired pneumonia: a systematic review and meta-analysis. BMC Infectious Diseases 2019;19(1):403. DOI: 10.1186/s12879-019-4048-6. [Context Link 1, 2] View abstract...
18. Viasus D, Gaia V, Manzur-Barbur C, Carratala J. Legionnaires' disease: update on diagnosis and treatment. Infectious Diseases and Therapy 2022;11(3):973-986. DOI: 10.1007/s40121-022-00635-7. [Context Link 1, 2, 3, 4, 5, 6] View abstract...
19. Uyeki TM, et al. Clinical practice guidelines by the Infectious Diseases Society of America: 2018 update on diagnosis, treatment, chemoprophylaxis, and institutional outbreak management of seasonal influenza. Clinical Infectious Diseases 2019;68(6):e1- e7. DOI: 10.1093/cid/ciy866. (Reaffirmed 2023 Apr) [Context Link 1, 2, 3] View abstract...
20. Evans SE, et al. Nucleic acid-based testing for noninfluenza viral pathogens in adults with suspected community-acquired pneumonia. An Official American Thoracic Society Clinical Practice Guideline. American Journal of Respiratory and Critical Care Medicine 2021;203(9):1070-1087. DOI: 10.1164/rccm.202102-0498ST. (Reaffirmed 2022 Aug) [Context Link 1, 2, 3] View abstract...
21. Bartley PS, et al. Bacterial coinfection in influenza pneumonia: Rates, pathogens, and outcomes. Infection Control and Hospital Epidemiology 2022;43(2):212-217. DOI: 10.1017/ice.2021.96. [Context Link 1, 2, 3, 4, 5] View abstract...
22. Deshpande A, et al. Influenza testing and treatment among patients hospitalized with community-acquired neumonia. Chest 2022;Online. DOI: 10.1016/j.chest.2022.01.053. [Context Link 1, 2] View abstract...
23. Antibacterial drugs for community-acquired pneumonia. Medical Letter on Drugs and Therapeutics 2021;63(1616):10-14. [Context Link 1, 2] View abstract...
24. Nair GB, Niederman MS. Updates on community acquired pneumonia management in the ICU. Pharmacology & Therapeutics 2021;217:Online. DOI: 10.1016/j.pharmthera.2020.107663. [Context Link 1, 2, 3] View abstract...
25. Chalmers JD, et al. Validation of the Infectious Diseases Society of America/American Thoratic Society minor criteria for intensive care unit admission in community-acquired pneumonia patients without major criteria or contraindications to intensive care unit care. Clinical Infectious Diseases 2011;53(6):503-511. DOI: 10.1093/cid/cir463. [Context Link 1] View abstract...
26. Webb BJ, et al. Antibiotic use and outcomes after implementation of the drug resistance in pneumonia score in ED patients with community-onset pneumonia. Chest 2019;156(5):843-851. DOI: 10.1016/j.chest.2019.04.093. [Context Link 1, 2, 3, 4] View abstract...
27. Niederman MS, Torres A. Severe community-acquired pneumonia. European Respiratory Review 2022;31(166):Online. DOI: 10.1183/16000617.0123-2022. [Context Link 1, 2, 3] View abstract...
28. Jain S, et al. Community-acquired pneumonia requiring hospitalization among U.S. adults. New England Journal of Medicine 2015;373(5):415-427. DOI: 10.1056/NEJMoa1500245. [Context Link 1, 2] View abstract...
29. Peyrani P, et al. Incidence and mortality of adults hospitalized with community-acquired pneumonia according to clinical course. Chest 2020;157(1):34-41. DOI: 10.1016/j.chest.2019.09.022. [Context Link 1, 2] View abstract...
30. Seo H, et al. Clinical relevance of necrotizing change in patients with community-acquired pneumonia. Respirology 2017;22(3):551-558. DOI: 10.1111/resp.12943. [Context Link 1] View abstract...
31. Parente DM, Cunha CB, Mylonakis E, Timbrook TT. The clinical utility of methicillin resistant Staphylococcus aureus (MRSA) nasal screening to rule out MRSA pneumonia: a diagnostic meta-analysis with antimicrobial stewardship implications. Clinical Infectious Diseases 2018;67(1):1-7. DOI: 10.1093/cid/ciy024. [Context Link 1, 2] View abstract...
32. Babbel D, Sutton J, Rose R, Yarbrough P, Spivak ES. Application of the DRIP score at a veterans affairs hospital. Antimicrobial Agents and Chemotherapy 2018;62(3):e02277-17. DOI: 10.1128/AAC.02277-17. [Context Link 1, 2] View abstract...
33. Cilloniz C, Dominedo C, Torres A. Multidrug resistant gram-negative bacteria in community-acquired pneumonia. Critical Care 2019;23(1):79. DOI: 10.1186/s13054-019-2371-3. [Context Link 1] View abstract...
34. Qaseem A, Etxeandia-Ikobaltzeta I, Fitterman N, Williams JW, Kansagara D. Appropriate use of high-flow nasal oxygen in hospitalized patients for initial or postextubation management of acute respiratory failure: a clinical guideline from the American College of Physicians. Annals of Internal Medicine 2021;174(7):977-984. DOI: 10.7326/M20-7533. [Context Link 1] View abstract...
35. Yu Y, Liu W, Jiang HL, Mao B. Pneumonia is associated with increased mortality in hospitalized COPD patients: a systematic review and meta-analysis. Respiration 2021;100(1):64-76. DOI: 10.1159/000510615. [Context Link 1] View abstract...

36. Gonakoti S, Osifo IF. Protein-energy malnutrition increases mortality in patients hospitalized with bacterial pneumonia: a retrospective nationwide database analysis. Cureus 2021;13(1):e12645. DOI: 10.7759/cureus.12645. [Context Link 1] View abstract...
37. Saleem N, Kulkarni A, Snow TAC, Ambler G, Singer M, Arulkumaran N. Effect of corticosteroids on mortality and clinical cure in community-acquired pneumonia: a systematic review, meta-analysis, and meta-regression of randomized control trials. Chest 2023;163(3):484-497. DOI: 10.1016/j.chest.2022.08.2229. [Context Link 1, 2] View abstract...
38. Corticosteroids in community-acquired pneumonia. Medical Letter on Drugs and Therapeutics 2020;62(1589):7-8. [Context Link 1, 2] View abstract...
39. Dequin PF, et al. Hydrocortisone in severe community-acquired pneumonia. New England Journal of Medicine 2023;388(21):1931-1941. DOI: 10.1056/NEJMoa2215145. [Context Link 1, 2] View abstract...
40. Metersky ML, Ma A, Houck PM, Bratzler DW. Antibiotics for bacteremic pneumonia: improved outcomes with macrolides but not fluoroquinolones. Chest 2007;131(2):466-473. DOI: 10.1378/chest.06-1426. [Context Link 1, 2] View abstract...
41. Garin N, et al. Beta-lactam monotherapy vs beta-lactam-macrolide combination treatment in moderately severe community-acquired pneumonia: a randomized noninferiority trial. JAMA Internal Medicine 2014;174(12):1894-1901. DOI: 10.1001/jamainternmed.2014.4887. [Context Link 1, 2] View abstract...
42. Shorr AF, et al. Readmission following hospitalization for pneumonia: the impact of pneumonia type and its implication for hospitals. Clinical Infectious Diseases 2013;57(3):362-367. DOI: 10.1093/cid/cit254. [Context Link 1, 2] View abstract...
43. Tang VL, Halm EA, Fine MJ, Johnson CS, Anzueto A, Mortensen EM. Predictors of rehospitalization after admission for pneumonia in the Veterans Affairs Healthcare System. Journal of Hospital Medicine 2014;9(6):379-383. DOI: 10.1002/jhm.2184. [Context Link 1, 2] View abstract...
44. Ramirez JA, et al. Treatment of community-acquired pneumonia in immunocompromised adults: a consensus statement regarding initial strategies. Chest 2020;158(5):1896-1911. DOI: 10.1016/j.chest.2020.05.598. [Context Link 1, 2, 3, 4, 5] View abstract...
45. Gil R, Webb BJ. Strategies for prediction of drug-resistant pathogens and empiric antibiotic selection in community-acquired pneumonia. Current Opinion in Pulmonary Medicine 2020;26(3):249-259. DOI: 10.1097/MCP.0000000000000670. [Context Link 1] View abstract...
46. Kerneis S, Visseaux B, Armand-Lefevre L, Timsit JF. Molecular diagnostic methods for pneumonia: how can they be applied in practice? Current Opinion in Infectious Diseases 2021;34(2):118-125. DOI: 10.1097/QCO.0000000000000713. [Context Link 1] View abstract...
47. Ebell MH, Bentivegna M, Cai X, Hulme C, Kearney M. Accuracy of biomarkers for the diagnosis of adult community-acquired pneumonia: a meta-analysis. Academic Emergency Medicine 2020;27(3):195-206. DOI: 10.1111/acem.13889. [Context Link 1] View abstract...
48. MarketScan Database, 2012-2013 (Copyright @2012-2013 Truven Health Analytics Inc. All Rights Reserved.); proprietary health insurance data sources (2013-2014); and Medicare 100% Standard Analytical File (2013). [Context Link 1]
49. Preyde M, Brassard K. Evidence-based risk factors for adverse health outcomes in older patients after discharge home and assessment tools: a systematic review. Journal of Evidence-Based Social Work 2011;8(5):445-68. DOI: 10.1080/15433714.2011.542330. [Context Link 1] View abstract...
50. Billings J, Dixon J, Mijanovich T, Wennberg D. Case finding for patients at risk of readmission to hospital: development of algorithm to identify high risk patients. British Medical Journal 2006;333(7563):327. DOI: 10.1136/bmj.38870.657917.AE. [Context Link 1] View abstract...
51. van Walraven C, et al. Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community. Canadian Medical Association Journal 2010;182(6):551-7. DOI: 10.1503/cmaj.091117. [Context Link 1] View abstract...
52. Hasan O, et al. Hospital readmission in general medicine patients: a prediction model. Journal of General Internal Medicine 2010;25(3):211-9. DOI: 10.1007/s11606-009-1196-1. [Context Link 1] View abstract...
53. Billings J, Blunt I, Steventon A, Georghiou T, Lewis G, Bardsley M. Development of a predictive model to identify inpatients at risk of re-admission within 30 days of discharge (PARR-30). BMJ Open 2012;2(4):e001667. DOI: 10.1136/bmjopen-2012-001667. [Context Link 1] View abstract...
54. Jack BW, et al. A reengineered hospital discharge program to decrease rehospitalization: a randomized trial. Annals of Internal Medicine 2009;150(3):178-87. [Context Link 1] View abstract...
55. Woz S, et al. Gender as risk factor for 30 days post-discharge hospital utilisation: a secondary data analysis. BMJ Open 2012;2(2):e000428. DOI: 10.1136/bmjopen-2011-000428. [Context Link 1] View abstract...
56. Capelastegui A, et al. Predictors of short-term rehospitalization following discharge of patients hospitalized with community-acquired pneumonia. Chest 2009;136(4):1079-85. DOI: 10.1378/chest.08-2950. [Context Link 1, 2] View abstract...
57. Jasti H, Mortensen EM, Obrosky DS, Kapoor WN, Fine MJ. Causes and risk factors for rehospitalization of patients hospitalized with community-acquired pneumonia. Clinical Infectious Diseases 2008;46(4):550-6. DOI: 10.1086/526526. [Context Link 1, 2] View abstract...
58. Aliberti S, et al. Association between time to clinical stability and outcomes after discharge in hospitalized patients with community-acquired pneumonia. Chest 2011;140(2):482-8. DOI: 10.1378/chest.10-2895. [Context Link 1, 2] View abstract...
59. Jencks SF, Williams MV, Coleman EA. Rehospitalizations among patients in the Medicare fee-for-service program. New England Journal of Medicine 2009;360(14):1418-28. DOI: 10.1056/NEJMsa0803563. [Context Link 1, 2] View abstract...
60. Arbaje AI, Wolff JL, Yu Q, Powe NR, Anderson GF, Boult C. Postdischarge environmental and socioeconomic factors and the likelihood of early hospital readmission among community-dwelling Medicare beneficiaries. Gerontologist 2008;48(4):495-504. [Context Link 1] View abstract...
61. Garcia-Perez L, Linertova R, Lorenzo-Riera A, Vazquez-Diaz JR, Duque-Gonzalez B, Sarria-Santamera A. Risk factors for hospital readmissions in elderly patients: a systematic review. Quarterly Journal of Medicine 2011;104(8):639-51. DOI: 10.1093/qjmed/hcr070. [Context Link 1] View abstract...
62. Silverstein MD, Qin H, Mercer SQ, Fong J, Haydar Z. Risk factors for 30-day hospital readmission in patients =65 years of age. Proceedings (Baylor University. Medical Center) 2008;21(4):363-72. [Context Link 1] View abstract...

63. Respiratory disorders. In: Nettina SM, editor. Lippincott Manual of Nursing Practice. 11th ed. Philadelphia: Wolters Kluwer Health | Lippincott Williams & Wilkins; 2019:191-231. [Context Link 1]
64. Mondor EE. Lower respiratory problems. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:596-631. [Context Link 1]
65. Hudson T. The role of social determinates of health in discharge practices. Nursing Clinics of North America 2021;56(3):369-378. DOI: 10.1016/j.cnur.2021.04.004. [Context Link 1, 2] View abstract...
66. Mayer RS, Noles A, Vinh D. Determination of postacute hospitalization level of care. Medical Clinics of North America 2020;104(2):345-357. DOI: 10.1016/j.mcna.2019.10.011. [Context Link 1] View abstract...
67. Centers for Medicare and Medicaid Services. "Condition of participation: Discharge planning." 42 CFR Pt. 482.43. Washington, DC 2022 Oct [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1, 2, 3]
68. Olson G, Davis AM. Diagnosis and treatment of adults with community-acquired pneumonia. Journal of the American Medical Association 2020;323(9):885-886. DOI: 10.1001/jama.2019.21118. [Context Link 1] View abstract...
69. Adverse drug reactions and medication errors. In: Burchum JR, Rosenthal LD, editors. Lehne's Pharmacology for Nursing Care. 11th ed. Elsevier; 2022:63-73. [Context Link 1, 2]
70. Cook LK, Wulf JA. CE: Community-acquired pneumonia: a review of current diagnostic criteria and management. American Journal of Nursing 2020;120(12):34-42. DOI: 10.1097/01.NAJ.0000723420.30838.97. [Context Link 1, 2, 3, 4] View abstract...
71. Saleeby J. Communication and collaboration. In: Perry AG, Potter PA, Ostendorf WR, editors. Nursing Interventions and Clinical Skills. 7th ed. Elsevier; 2020:9-21. [Context Link 1, 2]
72. Elements of Excellence in Transitions of Care (TOC). TOC Checklist [Internet] National Transitions of Care Coalition. Accessed at: https://www.ntocc.org. [accessed 2023 Aug 10] [Context Link 1]
73. Medical-surgical nursing. In: Hinkle JL, Cheever KH, Overbaugh KJ, editors. Brunner & Suddarth's Textbook of Medical-Surgical Nursing. 15th ed. Wolters Kluwer; 2022:33-55. [Context Link 1]
74. Transitions of care. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:137-144. [Context Link 1]
75. Becker C, et al. Interventions to improve communication at hospital discharge and rates of readmission: a systematic review and meta-analysis. JAMA Network Open 2021;4(8):e2119346. DOI: 10.1001/jamanetworkopen.2021.19346. [Context Link 1] View abstract...
76. The nursing process in drug therapy and patient safety. In: Tucker RG, editor. Karch's Focus on Nursing Pharmacology. 9th ed. Wolters Kluwer; 2023:47-57. [Context Link 1]
77. Ostendorf WR. Preparation for safe medication administration. In: Perry AG, Potter PA, Ostendorf WR, editors. Nursing Interventions and Clinical Skills. 7th ed. Elsevier; 2020:551-567. [Context Link 1]
78. The case manager's roles, functions, and activities. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:9-14. [Context Link 1]
79. Esden J. Treatment update: Outpatient management of community-acquired pneumonia. Nurse Practitioner 2020;45(3):16-25. DOI: 10.1097/01.NPR.0000653944.99226.25. [Context Link 1, 2, 3, 4] View abstract...
80. Pulmonary care plans. In: Gulanick M, Myers JL, editors. Nursing Care Plans. 10th ed. Elsevier; 2022:385-472. [Context Link 1]
81. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1]
82. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2023 Jun 05 [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1]
83. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 1 - Inpatient Hospital Services Covered Under Part A Rev. 10892 [Internet] Centers for Medicare and Medicaid Services. 2021 Aug Accessed at: https://www.cms.gov/manuals/. [accessed 2023 Sep 08] [Context Link 1]
84. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 6 - Hospital Services Covered Under Part B Rev. 10541 [Internet] Centers for Medicare and Medicaid Services. 2020 Dec Accessed at: http://www.cms.gov/manuals/. [accessed 2023 Nov 09] [Context Link 1]
85. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12171 [Internet] Centers for Medicare and Medicaid Services. 2023 Aug 03 Accessed at: https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/. [accessed 2023 Oct 25] [Context Link 1]
86. Centers for Medicare and Medicaid Services. Medicare Program Integrity Manual. Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment Rev. 10365 [Internet] Centers for Medicare and Medicaid Services. 2020 Oct 02 Accessed at: https://www.cms.gov/regulations-and-guidance/regulations-and-guidance. [accessed 2023 Oct 20] [Context Link 1]
87. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: https://www.cms.gov/medicare-coverage-database/search.aspx? Updated 2023 [accessed 2023 Oct 20] [Context Link 1]

Footnotes

[A] For patients with pneumonia suspected or known to be of viral etiology other than COVID-19 (eg, positive test for influenza), the Viral Illness, Acute ISC guideline is appropriate. For viral infection due to COVID-19, the COVID-19 ISC guideline should be used. If the etiology is bacterial or unknown (eg, treating with antibiotics empirically), the Pneumonia guideline is appropriate. [A in Context Link 1]

[B] This guideline pertains to the vast majority of patients with pneumonia, including patients deemed to have healthcare-associated pneumonia (HCAP). HCAP is defined as pneumonia developing in a patient with one or more of the following healthcare contacts: residence in a nursing home or long-term care facility, wound care, IV chemotherapy or hemodialysis within the previous 30 days, or acute care hospitalization for 2 or more days within the previous 90 days.(1) Concerns have been raised as to whether a patient identified as having HCAP is truly more likely to be infected with a multidrug-resistant (MDR) organism.(2)(3)(4) [B in Context Link 1, 2]

[C] An evidence-based specialty society guideline recommends that a clinical prediction rule for prognosis be used in addition to clinical judgment to determine inpatient vs outpatient treatment of community-acquired pneumonia.(2) This same guideline recommends use of the Pneumonia Severity Index (PSI) rather than the CURB-65.(2) A second guideline notes that the PSI is superior for identifying low-risk patients who may be appropriate for outpatient care.(6) [C in Context Link 1]

[D] Specialty guidelines recommend ICU admission for a patient with need for invasive or noninvasive assisted ventilation, Hemodynamic instability, or any 3 of the following severity factors: sodium less than 130 mEq/L (mmol/L), PaO_2/FiO_2 ratio of 250 mm Hg or less, multilobed infiltrates, Altered mental status, BUN 20 mg/dL (7.1 mmol/L) or greater, WBC count less than 4000/mm^3 (4 x10^9/L), platelet count less than 100,000/mm^3 (100 x10^9/L), or temperature less than 96.8 degrees F (36 degrees C). See Intensive, Intermediate, and Telemetry Care Guidelines ISC for further information.(2)(6)(24)(25) [D in Context Link 1]

[E] See Clinical Indications for Admission to Inpatient Care in this guideline. [E in Context Link 1]

[F] The patient may be discharged on oral antibiotics.(2) [F in Context Link 1]

[G] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [G in Context Link 1]

[H] The most useful and powerful risk factors for infection with antibiotic-resistant organisms are those that are locally developed and validated, as prevalence can vary significantly.(2) Concerning methicillin-resistant *Staphylococcus aureus* (MRSA) and *Pseudomonas* species, an evidence-based specialty society guideline concludes that the most consistently strong individual risk factors, other than those developed locally, are a patient having had either organism previously isolated, especially from the respiratory tract, or recent (within past 90 days) hospitalization with exposure to parenteral antibiotics.(2) This same guideline recommends that, when empiric coverage for *Pseudomonas* species or MRSA is used, therapy should be de-escalated (reversion to routine antibiotic coverage) if microbiological results (eg, blood culture, sputum culture) do not confirm infection with a resistant organism within 48 hours.(2) [H in Context Link 1]

[I] In pneumonia patients, the absence of nasal methicillin-resistant *Staphylococcus aureus* (MRSA) colonization upon admission indicates a very low likelihood of MRSA infection. A systematic review and meta-analysis (22 studies, 5163 patients) examining nasal colonization screening found a pooled sensitivity for MRSA pneumonia of 71% and a pooled specificity of 90%.(2)(31) Assuming a 10% prevalence of MRSA pneumonia, this would translate to a positive predictive value (PPV) of 45% and a negative predictive value (NPV) of 97%, indicating that a negative screen can essentially rule out MRSA pneumonia.(2)(31) In settings with a lower prevalence of MRSA pneumonia, the NPV would be even higher. The PPV, on the other hand, does not allow this screening to rule in MRSA pneumonia, although it may yield a likelihood that rises above the treatment (eg, IV vancomycin) threshold. Two studies, encompassing 901 patients admitted for community-acquired pneumonia, reported that a Drug Resistance in Pneumonia (DRIP) score less than 4 had a 99% NPV for having a drug-resistant pathogen.(26)(32) Low PPV for a DRIP score 4 or greater (4.7% to 8.4%) in these studies indicates the need to continue using antibiotic stewardship and de-escalation practices in conjunction with this score after broad spectrum antibiotics are started.(26)(32) A calculator is available at https://www.mdcalc.com/drug-resistance-pneumonia-drip-score. [I in Context Link 1, 2]

[J] Specific testing performed will depend on the etiology of immunocompromise.(44) [J in Context Link 1]

[K] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(65) [K in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
  - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
    - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
    - Decreased urine output,[C] as indicated by **ALL** of the following:
      - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
      - Urine output less than 0.5 mL/kg/hr for 12 hours
  - Significant clinical findings, as indicated by **1 or more** of the following[B]:
    - Hemodynamic instability
    - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
    - Altered mental status

- Cardiac arrhythmias of immediate concern
- Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
- Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
- Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[D] or persists despite observation care
- Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

References

1. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
2. Weisbord SD, Palevsky PM. Prevention and management of acute kidney injury. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:940-977.e15.
3. Palevsky PM, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for acute kidney injury. American Journal of Kidney Diseases 2013;61(5):649-72. DOI: 10.1053/j.ajkd.2013.02.349.
4. Whelton PK, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation 2018;138(17):e484-e594. DOI: 10.1161/CIR.0000000000000596. (Reaffirmed 2023 Jan)
5. Flynn JT, et al. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. Pediatrics 2017;140(3):e20171904. DOI: 10.1542/peds.2017-1904. (Reaffirmed 2022 Aug)
6. Mirvis DM, Goldberger AL. Electrocardiography. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:141-174.e3.
7. Mount DB. Fluid and electrolyte disturbances. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:338-356.

Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
C. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
D. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute[A] kidney injury (stage 3),[B] as indicated by **1 or more** of the following[C](1)(2)(3):
  - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
  - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
  - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m^2 (0.59 mL/sec/1.73m^2) eGFR - Pediatric Calculator and **1 or more** of the following:
    - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
    - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
  - Decreased urine output,[D] as indicated by **ALL** of the following:
    - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
    - Oligoanuria, as indicated by **1 or more** of the following:
      - Urine output less than 0.3 mL/kg/hour for 24 hours
      - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
  - Initiation of renal replacement therapy (RRT) required(4)

References

1. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
2. Weisbord SD, Palevsky PM. Prevention and management of acute kidney injury. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:940-977.e15.
3. Palevsky PM, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for acute kidney injury. American Journal of Kidney Diseases 2013;61(5):649-72. DOI: 10.1053/j.ajkd.2013.02.349.
4. Gaudry S, Palevsky PM, Dreyfuss D. Extracorporeal kidney-replacement therapy for acute kidney injury. New England Journal of Medicine 2022;386(10):964-975. DOI: 10.1056/NEJMra2104090.

Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
4. Schor NF. Neurologic evaluation. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:3053-3063.e1.

Altered mental status that is severe or persistent

- Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.

3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
4. Schor NF. Neurologic evaluation. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:3053-3063.e1.

Bacteremia

- Bacteremia refers to blood culture isolation of a bacterial species that is likely to be pathologic and not a contaminant.(1)(2)

References

1. Owens TA, Fowler VG Jr, Pilkington EF III. Infective endocarditis. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1528-1536.
2. Lowy FD. Staphylococcal infections. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:1178-1188.

Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
    - Resuscitated ventricular fibrillation or cardiac arrest
    - Ventricular escape rhythm
    - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
    - Nonsustained ventricular tachycardia and **1 or more** of the following:
      - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
      - Acute myocarditis
  - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
    - Type II second-degree atrioventricular block
    - Third-degree atrioventricular block
    - New-onset left bundle branch block with suspected myocardial ischemia
  - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
    - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
    - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
  - Heart rhythms of concern due to **1 or more** of the following:
    - Hypotension
    - Respiratory distress
    - Association with other significant symptoms (eg, syncope)(9)(10)(12)

References

1. Curtis AB, Tomaselli GF. Approach to the patient with cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1145-1162.e7.
2. Dalal AS, Van Hare GF. Disturbances of rate and rhythm of the heart. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:2436-2447.e1.
3. Joglar JA, et al. 2023 HRS Expert Consensus Statement on the management of arrhythmias during pregnancy. Heart Rhythm 2023;20(10):Online. DOI: 10.1016/j.hrthm.2023.05.017.
4. El-Chami M, Fernando L. Ventricular arrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1003-1014.
5. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2018;138(13):e272-e391. DOI: 10.1161/CIR.0000000000000549. (Reaffirmed 2023 Jul)
6. Epstein AE, et al. 2012 ACCF/AHA/HRS focused update incorporated into the ACCF/AHA/HRS 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. Circulation 2013;127(3):e283-e352. DOI: 10.1161/CIR.0b013e318276ce9b. (Reaffirmed 2022 Jun)
7. Miller JM, Ellenbogen KA. Therapy for cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1208-1244.
8. Park DS, Fishman GI. The cardiac conduction system. Circulation 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
9. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. Journal of the American College of Cardiology 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)

10. Chun EB, McGorisk GM. Supraventricular tachyarrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:980-995.
11. Hoskins MH, De Lurgio DB. Pacemakers, defibrillators, and cardiac resynchronization devices in hospital medicine. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1025-34.
12. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.

Care Task List

- Pneumonia care tasks to consider as soon as feasible may include:
  - Review results and care implications as appropriate:
    - Culture results
    - Repeat WBC count
    - Other ordered investigations
  - Schedule, facilitate, or confirm not needed as appropriate:
    - CT scan
    - Echocardiogram
    - Lung ventilation perfusion scan
    - Treatment and comorbidity investigation:
      - Thoracentesis
      - Bronchoscopy
      - IV access placement for any outpatient treatment
  - Advance toward planned diet.
  - Test saturation off oxygen and discontinue as appropriate.
  - Advance activity:
    - Out of bed
    - Ambulate short distance, with assistance as needed.
    - Advance to independent activity as tolerated.
    - Deficit assessment by physical therapist, if needed, identifying any discharge support requirements
  - Order as appropriate:
    - Antibiotic regimen revision (per culture results, patient response, GFR)
    - IV medication discontinuation, or adjustment instructions
    - IV fluid discontinuation
    - Conversion of all medications to oral route (or specification of outpatient IV antibiotic regimen)
  - Confirm patient status:
    - Volume status stable on oral intake
    - Vital signs, mental status, and respiratory function acceptable and stable or improving
    - Ambulatory or at acceptable activity for next level of care without significant dyspnea

Clinically active atrial fibrillation

- Clinically active atrial fibrillation refers to atrial fibrillation necessitating electrical cardioversion or treatment while hospitalized with parenteral (IV) medication to control heart rate (eg, digoxin, diltiazem, metoprolol) or to cardiovert (eg, amiodarone, procainamide).

Clinically active chronic obstructive pulmonary disease

- Clinically active chronic obstructive pulmonary disease (COPD) refers to COPD necessitating assisted ventilation (eg, BPAP, CPAP) or treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active diabetes

- Clinically active diabetes refers to diabetes necessitating treatment with insulin while hospitalized.

Clinically active heart failure

- Clinically active heart failure refers to heart failure necessitating treatment with parenteral (IV) diuretics (eg, furosemide) during first 1 to 2 days of hospital care.

Clinically significant dementia

- Clinically significant dementia refers to dementia necessitating treatment while hospitalized with antipsychotic medication (eg, haloperidol, risperidone) or dementia-specific medication (eg, donepezil, memantine).

Clinically significant malnutrition

- Clinically significant malnutrition refers to patients presenting with findings indicative of poor nutritional status; examples include unintentional weight loss, BMI less than 18.5, inadequate nutritional intake, loss of subcutaneous fat (eg, orbital, triceps, over ribs), or loss of muscle mass (eg, temples, clavicles, hands). BMI Calculator

Dehydration that is severe or persistent

- Dehydration that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Clinical findings of severe dehydration, as indicated by **1 or more** of the following:
    - Hemodynamic instability
    - Acute renal failure (stage 3 acute kidney injury)
    - Acute kidney injury (stage 2)
    - Serum sodium greater than 150 mEq/L (mmol/L)
  - Dehydration that is persistent, as indicated by **ALL** of the following:
    - Oral rehydration therapy not tolerated or insufficient to adequately correct dehydration
    - Appropriate intravenous treatment (eg, fluids) does not readily correct dehydration.

References

1. Padilipsky P, White W. Pediatric infectious diarrheal disease and dehydration. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2151-2168.e3.
2. Mount DB. Fluid and electrolyte disturbances. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:338-356.
3. Greenbaum LA. Deficit therapy. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:429-432.e1.
4. Greenbaum LA. Maintenance and replacement therapy. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:425-428.e1.

Extended Stay Risk Assessment

- Risk of extended length of stay is increased by presence of **1 or more** of the following(1):
  - Acute heart failure (eg, pulmonary edema)
  - Acute renal failure (stage 3 acute kidney injury)
  - Comorbid acute exacerbation of COPD

References

1. Premier PINC AI™ Healthcare Database (PHD), 01/01/2021-12/31/2022. Premier, Inc.

Fever

- Fever is defined as a core[A] temperature greater than or equal to 100.4 degrees F (38 degrees C).[B](2)(3)

References

1. Niven DJ, Gaudet JE, Laupland KB, Mrklas KJ, Roberts DJ, Stelfox HT. Accuracy of peripheral thermometers for estimating temperature: a systematic review and meta-analysis. Annals of Internal Medicine 2015;163(10):768-77. DOI: 10.7326/M15-1150.
2. Nield LS, Kamat D. Fever. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:1386-1388.e1.
3. Miller CS, Wiese JG. Hyperthermia and fever. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:647-656.

Footnotes

A. While a rectal temperature can be considered a core reading, it is not always practical. Tympanic membrane temperature is not usually considered a core reading; however, the devices that measure this temperature often are (or can be) programmed such that they automatically adjust readings to reflect a core temperature equivalent.(1)

B. There is not a universally accepted definition of fever, as various temperatures are proposed as the cut-off that constitutes a fever.

Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
  - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
    - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
    - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
    - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
  - Hypotension that is severe, as indicated by **ALL** of the following:

- Hypotension
- Severe hypotension, as indicated by **1 or more** of the following:
  - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
  - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
  - Mean arterial pressure[A][D] less than 65 mm Hg
  - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
3. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.
4. Turner DA, Cheifetz IM. Shock. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:572-583.e1.
5. Nyhan A. Cardiology. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:145-188.e4.
6. Naidu SS, et al. SCAI SHOCK stage classification expert consensus update: a review and incorporation of validation studies: this statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. Journal of the American College of Cardiology 2022;Online. DOI: 10.1016/j.jacc.2022.01.018.
7. Wardi G, Brice J, Correia M, Liu D, Self M, Tainter C. Demystifying lactate in the emergency department. Annals of Emergency Medicine 2020;75(2):287-298. DOI: 10.1016/j.annemergmed.2019.06.027.
8. Levitt DG, Levitt JE, Levitt MD. Quantitative assessment of blood lactate in shock: measure of hypoxia or beneficial energy source. BioMed Research International 2020;2020:2608318. DOI: 10.1155/2020/2608318.
9. Bakker J, Postelnicu R, Mukherjee V. Lactate: where are we now? Critical Care Clinics 2020;36(1):115-124. DOI: 10.1016/j.ccc.2019.08.009.
10. Kraut JA, Madias NE. Lactic acidosis. New England Journal of Medicine 2014;371(24):2309-2319. DOI: 10.1056/NEJMra1309483.
11. Olivieri L, Chasm R. Diabetic ketoacidosis in the pediatric emergency department. Emergency Medicine Clinics of North America 2013;31(3):755-773. DOI: 10.1016/j.emc.2013.05.004.
12. Maloney GE, Glauser JM. Diabetes mellitus and disorders of glucose homeostasis. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1543-1558.e2.
13. Arnold TD, Miller M, van Wessem KP, Evans JA, Balogh ZJ. Base deficit from the first peripheral venous sample: a surrogate for arterial base deficit in the trauma bay. Journal of Trauma 2011;71(4):793-7; discussion 797. DOI: 10.1097/TA.0b013e31822ad694.
14. Malinoski DJ, Todd SR, Slone S, Mullins RJ, Schreiber MA. Correlation of central venous and arterial blood gas measurements in mechanically ventilated trauma patients. Archives of Surgery 2005;140(11):1122-5. DOI: 10.1001/archsurg.140.11.1122.
15. Zakrison T, McFarlan A, Wu YY, Keshet I, Nathens A. Venous and arterial base deficits: do these agree in occult shock and in the elderly? A Bland-Altman analysis. Journal of Trauma and Acute Care Surgery 2013;74(3):936-9. DOI: 10.1097/TA.0b013e318282747a.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated.(12)
D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals 1/3 SBP + 2/3 DBP.(1)(4) A calculator is available at https://www.mdcalc.com/mean-arterial-pressure-map.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
  - Hemodynamic abnormalities at baseline or acceptable for next level of care
  - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
3. Ingbar DH, Thiele H. Cardiogenic shock and pulmonary edema. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2250-2257.
4. Brant EB, Seymour CW, Angus DC. Sepsis and septic shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2241-2249.
5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
  - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
  - Low blood pressure, as indicated by **1 or more** of the following:
    - SBP less than 90 mm Hg in adult or child 10 years or older[A]
    - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
    - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
    - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
    - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
2. Massaro AF. Approach to the patient with shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2235-2241.
3. Horeczko T. Pediatric cardiac disorders. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2109-2131.e1.
4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at https://www.mdcalc.com/mean-arterial-pressure-map.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
  - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older

- Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
- SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
- SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
2. Massaro AF. Approach to the patient with shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2235-2241.
3. Horeczko T. Pediatric cardiac disorders. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2109-2131.e1.
4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
  - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO_2) of less than 90% or arterial blood gas partial pressure of oxygen (PO_2) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
  - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO_2 greater than 89% or PO_2 greater than 59 mm Hg (7.9 kPa)[A]
  - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

1. Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
2. Jamali H, et al. Racial disparity in oxygen saturation measurements by pulse oximetry: evidence and implications. Annals of the American Thoracic Society 2022;19(12):1951-1964. DOI: 10.1513/AnnalsATS.202203-270CME.
3. Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

A. Pulse oximetry (SpO2) may underestimate arterial saturation (SaO2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Hypoxemia absent

- Hypoxemia absent, as indicated by **1 or more** of the following(1):

- Arterial oxygen saturation (SaO_2) of 90% or greater or arterial partial pressure of oxygen (PO_2) of 60 mm Hg (8.0 kPa) or greater on room air[A][B]
- Oxygen requirements are stable and arranged for at next level of care.

References

1. Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
2. Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

A. Specific target values (ie, 90% or 60 mm Hg (8.0 kPa)) may require adjustment when care is delivered at high altitude.(2)
B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Immunosuppressed

- Immunosuppressed refers to a degree of immunocompromise that is clinically significant (eg, affects choice of antimicrobial agent due to possibility of atypical organism, necessitates additional isolation, or leads to clinically relevant increase in likelihood of severe infection)[A]; examples include(1)(2)(3)(4)(5):
  - Neutropenia (absolute neutrophil count less than 500/mm^3 (0.5 x10^9/L), or expected to decrease to this level within the next 48 hours)
  - Drug-induced immunosuppression (ie, current usage); examples include:
    - Systemic corticosteroids[B](6)(7)(8)(9)
    - Immunosuppressive agent (eg, tacrolimus, azathioprine, methotrexate, cyclosporin)
    - Immunomodulatory therapy (eg, TNF inhibitor, interleukin inhibitor, monoclonal antibodies)
  - Poorly controlled diabetes mellitus (eg, HbA1c of 8.5% (69 mmol/mol) or higher)[C][D](10)(11)(12)(13)
  - Chronic kidney disease (ie, estimated GFR less than 60 mL/min/1.73m^2 (1.0 mL/sec/1.73m^2)) or end-stage renal disease (hemodialysis or peritoneal dialysis)[E](15)(16)(17)(18)(19)(20)
  - Cirrhosis[F](21)(22)(23)(24)(25)
  - Bone marrow suppression or dysfunction (eg, chemotherapy, radiation therapy, bone marrow fibrosis or infiltration with malignant cells)
  - Humoral (antibody, B-lymphocyte) immune dysfunction or deficiency (eg, common variable immune deficiency, specific antibody deficiency, chronic lymphocytic leukemia)(26)(27)
  - Cell-mediated (eg, T-lymphocyte, natural killer cell) immune dysfunction or deficiency (eg, severe combined immune deficiency, DiGeorge syndrome, Wiskott-Aldrich syndrome)
  - Phagocytic cell (neutrophil, macrophage) dysfunction (eg, chronic granulomatous disease, leukocyte adhesion deficiency)
  - Acquired immunodeficiencies (eg, HIV/AIDS, hematopoietic and lymphoid malignancies)
  - History of solid organ or hematopoietic stem cell transplant[G]
  - Asplenia (surgical or congenital) or functional hyposplenia (eg, sickle cell disease)[H](28)
  - Complement deficiency (eg, hereditary, due to protein loss (nephrotic syndrome))

References

1. Holland SM, Gallin JI. Evaluation of the patient with suspected immunodeficiency. In: Bennett JE, Dolin R, Blaser MJ, editors. Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases. 9th ed. Philadelphia, PA: Elsevier; 2020:142-153.e2.
2. Perkins J, Waasdorp CP. The immunocompromised patient. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2348-2360.e2.
3. Sifri CD, Marr KA. Approach to fever and suspected infection in the immunocompromised host. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:1815-1825.e2.
4. Infections of the Immunocompromised Host. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1646-1652.
5. Michaels MG, Chong HJ, Green M. Infections in immunocompromised persons. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:1403-1410.e2.
6. Youssef J, Novosad SA, Winthrop KL. Infection risk and safety of corticosteroid use. Rheumatic Diseases Clinics of North America 2016;42(1):157-176, ix-x. DOI: 10.1016/j.rdc.2015.08.004.
7. George MD, et al. Risk for serious infection with low-dose glucocorticoids in patients with rheumatoid arthritis : a cohort study. Annals of Internal Medicine 2020;173(11):870-878. DOI: 10.7326/M20-1594.

8. Wu J, Keeley A, Mallen C, Morgan AW, Pujades-Rodriguez M. Incidence of infections associated with oral glucocorticoid dose in people diagnosed with polymyalgia rheumatica or giant cell arteritis: a cohort study in England. Canadian Medical Association Journal 2019;191(25):E680-E688. DOI: 10.1503/cmaj.190178.
9. Ramirez JA, et al. Treatment of community-acquired pneumonia in immunocompromised adults: a consensus statement regarding initial strategies. Chest 2020;158(5):1896-1911. DOI: 10.1016/j.chest.2020.05.598.
10. Hine JL, et al. Association between glycaemic control and common infections in people with Type 2 diabetes: a cohort study. Diabetic Medicine 2017;34(4):551-557. DOI: 10.1111/dme.13205.
11. Critchley JA, Carey IM, Harris T, DeWilde S, Hosking FJ, Cook DG. Glycemic control and risk of infections among people with type 1 or type 2 diabetes in a large primary care cohort study. Diabetes Care 2018;41(10):2127-2135. DOI: 10.2337/dc18-0287.
12. Mor A, Dekkers OM, Nielsen JS, Beck-Nielsen H, Sorensen HT, Thomsen RW. Impact of glycemic control on risk of infections in patients with type 2 diabetes: a population-based cohort study. American Journal of Epidemiology 2017;186(2):227-236. DOI: 10.1093/aje/kwx049.
13. Luk AOY, et al. Glycaemia control and the risk of hospitalisation for infection in patients with type 2 diabetes: Hong Kong Diabetes Registry. Diabetes/Metabolism Research and Reviews 2017;33(8):Online. DOI: 10.1002/dmrr.2923.
14. American Diabetes Association. Standards of medical care in diabetes - 2023. Diabetes Care 2023;45(Supplement 1):S1-S291.
15. Ishigami J, Matsushita K. Clinical epidemiology of infectious disease among patients with chronic kidney disease. Clinical and Experimental Nephrology 2019;23(4):437-447. DOI: 10.1007/s10157-018-1641-8.
16. Ishigami J, Grams ME, Chang AR, Carrero JJ, Coresh J, Matsushita K. CKD and risk for hospitalization with infection: the atherosclerosis risk in communities (ARIC) study. American Journal of Kidney Diseases 2017;69(6):752-761. DOI: 10.1053/j.ajkd.2016.09.018.
17. Narayanan M. The many faces of infection in CKD: evolving paradigms, insights, and novel therapies. Advances in Chronic Kidney Disease 2019;26(1):5-7. DOI: 10.1053/j.ackd.2018.10.001.
18. Crepin T, et al. Uraemia-induced immune senescence and clinical outcomes in chronic kidney disease patients. Nephrology, Dialysis, Transplantation 2020;35(4):624-632. DOI: 10.1093/ndt/gfy276.
19. Yeun JY, Young B, Depner TA, Chin AA. Hemodialysis. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:2038-2093 e21.
20. Betjes MG. Immune cell dysfunction and inflammation in end-stage renal disease. Nature Reviews. Nephrology 2013;9(5):255-65. DOI: 10.1038/nrneph.2013.44.
21. Kamath PS, Shah VH. Overview of cirrhosis. In: Feldman M, Friedman LS, Brandt LJ, editors. Sleisenger & Fordtran's Gastrointestinal and Liver Disease. 11th ed. Philadelphia, PA: Elsevier; 2021:1164-1171.e1.
22. Zangeneh TT, El Ramahi R, Klotz SA. Bacterial and miscellaneous infections of the liver. In: Sanyal AJ, Boyer TD, Lindor KD, Terrault NA, editors. Zakim and Boyer's Hepatology. 7th ed. Philadelphia, PA 19103-2899: Elsevier; 2018:579-592.e4.
23. Piano S, Angeli P. Bacterial infections in cirrhosis as a cause or consequence of decompensation? Clinics in Liver Disease 2021;25(2):357-372. DOI: 10.1016/j.cld.2021.01.006.
24. Albillos A, Martin-Mateos R, Van der Merwe S, Wiest R, Jalan R, Alvarez-Mon M. Cirrhosis-associated immune dysfunction. Nature Reviews. Gastroenterology & Hepatology 2022;19(2):112-134. DOI: 10.1038/s41575-021-00520-7.
25. Campbell KA, Trivedi HD, Chopra S. Infections in cirrhosis: a guide for the clinician. American Journal of Medicine 2021;134(6):727-734. DOI: 10.1016/j.amjmed.2021.01.015.
26. Ameratunga R, Allan C, Woon ST. Defining common variable immunodeficiency disorders in 2020. Immunology and Allergy Clinics of North America 2020;40(3):403-420. DOI: 10.1016/j.iac.2020.03.001.
27. Perez EE, Ballow M. Diagnosis and management of specific antibody deficiency. Immunology and Allergy Clinics of North America 2020;40(3):499-510. DOI: 10.1016/j.iac.2020.03.005.
28. Squire JD, Sher M. Asplenia and hyposplenism: an underrecognized immune deficiency. Immunology and Allergy Clinics of North America 2020;40(3):471-483. DOI: 10.1016/j.iac.2020.03.006.

Footnotes

A. Examples are given for some of the mechanisms of immunocompromise; however, the list is not all-encompassing. The intent is to illustrate the degree of immunocompromise that may be clinically significant. The presence of a listed potentially immunocompromising factor does not always indicate clinically significant immunosuppression. For example, some findings, such as neutropenia, are usually indicative of significant immunocompromise, whereas others (eg, presence of diabetes, renal failure, or asplenia) may not necessarily indicate clinically important immunocompromise. The judgment as to the presence of clinically significant immunocompromise in a particular patient is also influenced by several variables, such as the nature of the suspected infection, and the specific details of the patient's presentation, condition, and comorbidities. Patients may have more than one potential source of immunocompromise, increasing the likelihood of significant dysfunction (eg, poorly controlled diabetes and chronic kidney disease).

B. The minimum dose of systemic corticosteroid use associated with an increased risk of serious infection is not defined, but there is increasing risk with higher doses and longer duration of treatment.(4) Observational studies of patients with rheumatologic disease indicated, after statistical adjustment, that infection risk was increased with prolonged use of as little as 5 mg of prednisone-equivalent per day dosing, and also found that higher doses and longer duration increased risk.(6)(7)(8) An evidence-based specialty society guideline defined clinically important immunocompromise in adult pneumonia patients as those patients with an impaired immune response who are judged to be at elevated risk of infection by atypical or opportunistic organisms, so that there is a need to alter empirical antimicrobial therapy.(9) This same guideline stated that corticosteroid use would lead to such an immune system impairment at doses of 20 mg prednisone-equivalent per day or more for 14 days or more, or a course of treatment with a cumulative dose of 600 mg prednisone-equivalent or more.(9)

C. It is generally agreed that the presence of diabetes is associated with an increased risk of infection, including infection leading to hospitalization.(2)(10)(11)(12)(13) At least some of this risk is attributed to the deleterious effects of diabetes on the immune system.(2)(10)(11)(12)(13) It has been found that the impact of diabetes can be higher in certain types of infections (eg, higher impact in cellulitis, osteomyelitis) and with a longer total duration of the diagnosis.(10)(11)(13)
D. There is no universally agreed upon HbA1c level that indicates "poor control." It has been found that the higher the HbA1c, the larger the impact on the immune system.(10)(11)(12)(13) Goal HbA1c can vary to some degree depending on the patient. Otherwise healthy adults and adolescents may have a target HbA1c of less than 7% (53 mmol/mol), whereas those who have complex comorbidities, impairment in cognition or activities of daily living, or a history of severe hypoglycemia may have a target of less than 8% (64 mmol/mol).(14) However, these targets are put forth with reduction of microvascular and macrovascular effects of diabetes in mind, not immune system effects.(14) In this way, there is no simple cutoff between adequate and poor control. Cohort studies, after statistical adjustment, have found that HbA1c levels above 8.0% to 8.5% (64 to 69 mmol/mol) are independently associated with hospitalization due to infection.(10)(11)(12)(13)
E. Patients with chronic kidney disease or end-stage renal disease (ESRD) have impaired immune system function, measured, for example, by higher rates of infection requiring hospitalization.(15)(16)(17) The degree of renal impairment correlates with the severity of immunocompromise.(15)(16)(17) The degree of proteinuria (eg, urine albumin-to-creatinine ratio) is also independently correlated with the severity of immune system dysfunction. These 2 factors are additive, in that patients with both reduced GFR and significant proteinuria are the most severely affected.(15)(16)(17) Patients with ESRD, in general, have more profound immune system dysfunction, with inadequate dialysis even more detrimental.(18)(19)(20)
F. Patients with cirrhosis of the liver are functionally immunocompromised due to a variety of mechanisms. This immunocompromise correlates with the severity of the cirrhosis and is called cirrhosis-associated immune dysfunction (CAID). Immune dysfunction is also worsened during periods of acute exacerbation of chronic liver disease.(21)(22)(23)(24)(25) CAID is particularly associated with bacterial infections, affecting both the frequency and severity of infection.(23)(24)(25)
G. Susceptibility to infection varies with the immunosuppressive agents used.(5)
H. Patients with asplenia or hyposplenism are at higher risk of infection with encapsulated organisms, such as *Streptococcus pneumoniae*, meningococcal infection, encapsulated strains of *Haemophilus influenzae*, and certain tick-borne diseases.(2)(28)

Observation care

- When a Clinical Indication says "…despite (or beyond) observation care," the intent is to signify that, despite an attempt to address a sign, symptom, finding, or treatment need in observation care, a continued need for hospital-based care persists beyond the observation care time frame (ie, Observation Care Discharge Criteria are not met). The time frame of observation care is determined by regulation (eg, CMS's Two-Midnight Rule) or payer-provider contractual agreement. MCG's understanding of observation care is that it must be finite and of short duration. The Two-Midnight Rule for Medicare patients (inpatient is defined as medical necessity for hospital-based care across 2 or more midnights) provides a good example of what is meant by a short duration (ie, observation care should not continue beyond the second midnight).

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
    - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
    - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

1. Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. Journal of the American Society of Hypertension 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
2. Dalal AS, Van Hare GF. Syncope. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:566-571.e1.
3. Fang JC, O'Gara PT. History and physical examination: an evidence-based approach. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:123-140.

Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for

deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Pneumonia Multiple Condition Management Guidelines

- For patients who also have:
  - Clinically active atrial fibrillation, see Pneumonia with Clinically Active Atrial Fibrillation MCM
  - Clinically active chronic obstructive pulmonary disease and Clinically active diabetes, see Pneumonia with Clinically Active Chronic Obstructive Pulmonary Disease and Clinically Active Diabetes MCM
  - Clinically active chronic obstructive pulmonary disease, see Pneumonia with Clinically Active Chronic Obstructive Pulmonary Disease MCM
  - Clinically active diabetes, see Pneumonia with Clinically Active Diabetes MCM
  - Clinically active heart failure, see Pneumonia with Clinically Active Heart Failure MCM
  - Clinically significant dementia, see Pneumonia with Clinically Significant Dementia MCM
  - Clinically significant malnutrition, see Pneumonia with Clinically Significant Malnutrition MCM
  - Severe renal failure, see Pneumonia with Severe Renal Failure MCM

Readmission Risk Assessment

- Risk of readmission is increased by presence of **1 or more** of the following(1)(2)(3)(4)(5)(6)(7)(8):
  - Hospitalization (nonelective) in past 3 months
  - 2 or more emergency department visits in past 6 months
  - Admission from long-term care facility (eg, nursing home)
  - Prescribed antibiotics within 30 days prior to admission
  - Immunosuppressed patient
  - No source of outpatient care other than emergency department (eg, no primary care provider)(9)(10)(11)(12)(13)
  - Severe care transition barriers (eg, no caregiver, homeless)(11)(12)(13)(14)(15)(16)
  - Severe or end-stage renal disease (on dialysis or GFR less than 30 mL/min/1.73m^2 (0.5 mL/sec/1.73m^2))(14)(17)

  eGFR - Adult Calculator

References

1. MarketScan Database, 2012-2013 (Copyright @2012-2013 Truven Health Analytics Inc. All Rights Reserved.); proprietary health insurance data sources (2013-2014); and Medicare 100% Standard Analytical File (2013).
2. Preyde M, Brassard K. Evidence-based risk factors for adverse health outcomes in older patients after discharge home and assessment tools: a systematic review. Journal of Evidence-Based Social Work 2011;8(5):445-68. DOI: 10.1080/15433714.2011.542330.
3. Billings J, Dixon J, Mijanovich T, Wennberg D. Case finding for patients at risk of readmission to hospital: development of algorithm to identify high risk patients. British Medical Journal 2006;333(7563):327. DOI: 10.1136/bmj.38870.657917.AE.
4. van Walraven C, et al. Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community. Canadian Medical Association Journal 2010;182(6):551-7. DOI: 10.1503/cmaj.091117.
5. Hasan O, et al. Hospital readmission in general medicine patients: a prediction model. Journal of General Internal Medicine 2010;25(3):211-9. DOI: 10.1007/s11606-009-1196-1.
6. Billings J, Blunt I, Steventon A, Georghiou T, Lewis G, Bardsley M. Development of a predictive model to identify inpatients at risk of re-admission within 30 days of discharge (PARR-30). BMJ Open 2012;2(4):e001667. DOI: 10.1136/bmjopen-2012-001667.
7. Shorr AF, et al. Readmission following hospitalization for pneumonia: the impact of pneumonia type and its implication for hospitals. Clinical Infectious Diseases 2013;57(3):362-367. DOI: 10.1093/cid/cit254.
8. Tang VL, Halm EA, Fine MJ, Johnson CS, Anzueto A, Mortensen EM. Predictors of rehospitalization after admission for pneumonia in the Veterans Affairs Healthcare System. Journal of Hospital Medicine 2014;9(6):379-383. DOI: 10.1002/jhm.2184.
9. Jack BW, et al. A reengineered hospital discharge program to decrease rehospitalization: a randomized trial. Annals of Internal Medicine 2009;150(3):178-87.
10. Woz S, et al. Gender as risk factor for 30 days post-discharge hospital utilisation: a secondary data analysis. BMJ Open 2012;2(2):e000428. DOI: 10.1136/bmjopen-2011-000428.
11. Capelastegui A, et al. Predictors of short-term rehospitalization following discharge of patients hospitalized with community-acquired pneumonia. Chest 2009;136(4):1079-85. DOI: 10.1378/chest.08-2950.
12. Jasti H, Mortensen EM, Obrosky DS, Kapoor WN, Fine MJ. Causes and risk factors for rehospitalization of patients hospitalized with community-acquired pneumonia. Clinical Infectious Diseases 2008;46(4):550-6. DOI: 10.1086/526526.
13. Aliberti S, et al. Association between time to clinical stability and outcomes after discharge in hospitalized patients with community-acquired pneumonia. Chest 2011;140(2):482-8. DOI: 10.1378/chest.10-2895.
14. Jencks SF, Williams MV, Coleman EA. Rehospitalizations among patients in the Medicare fee-for-service program. New England Journal of Medicine 2009;360(14):1418-28. DOI: 10.1056/NEJMsa0803563.
15. Arbaje AI, Wolff JL, Yu Q, Powe NR, Anderson GF, Boult C. Postdischarge environmental and socioeconomic factors and the likelihood of early hospital readmission among community-dwelling Medicare beneficiaries. Gerontologist 2008;48(4):495-504.

16. Garcia-Perez L, Linertova R, Lorenzo-Riera A, Vazquez-Diaz JR, Duque-Gonzalez B, Sarria-Santamera A. Risk factors for hospital readmissions in elderly patients: a systematic review. Quarterly Journal of Medicine 2011;104(8):639-51. DOI: 10.1093/qjmed/hcr070.
17. Silverstein MD, Qin H, Mercer SQ, Fong J, Haydar Z. Risk factors for 30-day hospital readmission in patients =65 years of age. Proceedings (Baylor University. Medical Center) 2008;21(4):363-72.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
  - Patient with **1 or more** of the following:
    - Dyspnea (difficulty breathing)
    - Tachypnea
    - Abnormal breathing pattern (eg, chest retractions)
    - Other evidence of difficulty breathing
  - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
    - Hypoxemia
    - Altered mental status
    - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Severe renal failure

- Severe renal failure refers to renal failure or insufficiency necessitating inpatient dialysis or treatment with phosphate-binding agent medication (eg, sevelamer), or medication to treat secondary hyperparathyroidism (eg, cinacalcet).

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
  - Housing insecurity, as indicated by **1 or more** of the following:
    - Individual or caregiver's current living situation is **1 or more** of the following(5):
      - Does not have own housing (eg, staying in a hotel, shelter, or with others)
      - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
      - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
    - Current housing has **1 or more** of the following:
      - Electrical appliances (eg, stove, refrigerator) not working or unavailable
      - Insufficient heating or cooling
      - Insufficient ventilation
      - Lead paint or pipes
      - Mold
      - Pests (eg, bugs) or rodents
      - Smoke detectors not working or unavailable
  - Food insecurity, as indicated by **1 or more** of the following(6):
    - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
    - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
  - Insufficient transportation, as indicated by **1 or more** of the following(7):
    - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
    - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
  - Insufficient utilities, as indicated by **1 or more** of the following(8):
    - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
    - In the past year, electric, water, gas, or oil company threatened to shut off services.
  - Personal safety risk, as indicated by **2 or more** of the following(6):
    - Individual is sometimes or frequently physically hurt by another person (including family member).
    - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
    - Individual is sometimes or frequently threatened with physical harm by another person (including family member).

- Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
  - In the past year, individual or caregiver was unable to work due to lack of dependent care.
  - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
  - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
  - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
  - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
2. Moen M, Storr C, German D, Friedmann E, Johantgen M. A review of tools to screen for social determinants of health in the United States: a practice brief. Population Health Management 2020;23(6):422-429. DOI: 10.1089/pop.2019.0158.
3. Daniel-Robinson L, Moore JE. Innovation and Opportunities to Address Social Determinants of Health in Medicaid Managed Care. [Internet] Institute for Medicaid Innovation. 2019 Jan Accessed at: https://medicaidinnovation.org/emerging-topics/. [accessed 2023 Oct 20]
4. Billioux A, Verlander K, Anthony S, Alley D. Standardized Screening for Health-Related Social Needs in Clinical Settings: the Accountable Health Communities Screening Tool. [Internet] National Academy of Sciences. 2017 May Accessed at: https://nam.edu/. [accessed 2023 Sep 18]
5. Sandel M, et al. Unstable housing and caregiver and child health in renter families. Pediatrics 2018;14(2):e20172199. DOI: 10.1542/peds.2017-2199.
6. Children's HealthWatch Survey. Screening Instrument [Internet] Children's HealthWatch. 2020 Sep Accessed at: https://childrenshealthwatch.org/. [accessed 2023 Sep 11]
7. PRAPARE®: Protocol for Responding to and Assessing Patient Assets, Risks, and Experiences Screening Tool. [Internet] Association of Asian Pacific Community Health Organizations (AAPCHO) and National Association of Community Health Centers (NACHC). 2016 Sep Accessed at: https://prapare.org/the-prapare-screening-tool/. [accessed 2023 Sep 21]
8. Cook JT, et al. A brief indicator of household energy security: associations with food security, child health, and child development in US infants and toddlers. Pediatrics 2008;122(4):e867-75. DOI: 10.1542/peds.2008-0286.

Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
  - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
  - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
  - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
  - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
  - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
  - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

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Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
  - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
  - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
  - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
  - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
  - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
  - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
  - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Tachypnea absent

- Tachypnea[A][B] absent, as indicated by respiratory rate of **1 or more** of the following(2)(3):
  - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
  - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
  - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
  - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
  - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
  - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
  - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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Codes

ICD-10 Diagnosis: A21.2, A22.1, A37.01, A37.11, A37.81, A37.91, A48.1, A70, A78, B37.1, B44.0, J13, J14, J15.0, J15.1, J15.20, J15.211, J15.212, J15.29, J15.3, J15.4, J15.5, J15.61, J15.69, J15.7, J15.8, J15.9, J16.0, J16.8, J17, J18.0, J18.1, J18.8, J18.9 [Hide]

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